

Joint Alias–Parity–Shell Matching: Fixed-Reference Certificates and Exponential Contrast Precision

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Abstract

The first-alias ledger now has actual localized boundary, neighboring-shell, and far-remainder slots, but the shell is not identified with the earlier forward Gaussian laws. We formulate the exact fixed-reference matching equation without making that identification. If a typed shell representation has scale L , physical contrast c_{phys} , fixed deterministic contrast c_0 , and observation error $\mathcal{E}_{\text{obs}}$, then the full signed packet is exactly

$$e = L(c_{\text{phys}}^{2k} - y) + \mathcal{E}_{\text{obs}} + \mathcal{R}, \quad y = c_0^{2k} + \frac{\mathcal{A} - \mathcal{P} - \mathcal{B}}{L}.$$

In alias-normalized variables this gives $y = z_0 + (1 - q - b)/\ell$. On a fixed phase, the parity ratio and endpoint clearance remain synchronized. If L is comparable to the alias scale, y stays away from zero, and the observation error plus far remainder is $o(kR^{-2k})$, target closure forces contrast-radius precision $o((\beta R)^{-2k}/k)$. We also prove a sharp interval certificate retaining every trace-observation Duhamel weight, and an explicit scoped counterexample: the best-case reachability distance may vanish while the physical mismatch is order L and diverges relative to kR^{-2k} . The shell representation, contrast, observation bound, and far remainder remain unproved for the actual operator; no full-trace replacement or Riemann-hypothesis conclusion follows.

1 The actual packet and retained interface

Let $u_c \in (1, 2)$ solve $u^3 - 2u^2 + 2u - 2 = 0$, put

$$f(x) = 1 - u_c x^2, \quad \lambda = 2u_c(u_c - 1), \quad r_H = 0.85, \quad R = 1.4, \quad (1)$$

and let K_σ be the row-normalized folded noisy Markov operator on $[0, 1]$. For a measurable $J \subset [0, 1]$, let M_J be multiplication by its indicator and define the localized noisy and deterministic cyclic observations

$$L_{\sigma,n}(J) = \text{Tr}(M_J K_\sigma^n), \quad (2)$$

$$P_n(J) = \sum_{\substack{f^n(x)=x \\ x \in J}} \frac{1}{|1 - (f^n)'(x)|}. \quad (3)$$

We retain the RH-327 physical basepoint partition. If p_{2k} is the distinguished boundary periodic point, write

$$\Delta_k^{\text{clr}} = 1 - p_{2k} = C_b \lambda^{-2k} \{1 + o(1)\}, \quad C_b > 0, \quad (4)$$

$$h(x) = \sqrt{\frac{1 - \sqrt{(1-x)/u_c}}{u_c}}, \quad s_{\sigma,k} = h(p_{2k}), \quad (5)$$

and set

$$b_* = u_c^{-1/2}, \quad q_{b,\sigma,k} = \frac{b_* - s_{\sigma,k}}{\sqrt{\sigma}}, \quad q_{\sigma,k}(x) = \frac{x - s_{\sigma,k}}{\sqrt{\sigma}}. \quad (6)$$

For a fixed radius $A > 0$, the two half-open windows are

$$J_{\sigma,k,A}^- = \{x \in [0, 1] : q_{b,\sigma,k} - A \leq q_{\sigma,k}(x) < q_{b,\sigma,k}\}, \quad (7)$$

$$J_{\sigma,k,A}^+ = \{x \in [0, 1] : q_{b,\sigma,k} \leq q_{\sigma,k}(x) \leq q_{b,\sigma,k} + A\}, \quad (8)$$

and $F_{\sigma,k,A}$ is their complement. The actual Hardy-scaled slots at order $2k$ are

$$\mathcal{B}_{\sigma,k} = r_H^{-2k} \{L_{\sigma,2k}(J_{\sigma,k,A}^-) - P_{2k}(J_{\sigma,k,A}^-)\}, \quad (9)$$

$$\mathcal{S}_{\sigma,k} = r_H^{-2k} \{L_{\sigma,2k}(J_{\sigma,k,A}^+) - P_{2k}(J_{\sigma,k,A}^+)\}, \quad (10)$$

$$\mathcal{R}_{\sigma,k} = r_H^{-2k} \{L_{\sigma,2k}(F_{\sigma,k,A}) - P_{2k}(F_{\sigma,k,A})\}. \quad (11)$$

With

$$\mathcal{T}_{\sigma,2k} := r_H^{-2k} \{\text{Tr } K_{\sigma}^{2k} - P_{2k}([0, 1])\}, \quad (12)$$

they satisfy the exact actual partition $\mathcal{T}_{\sigma,2k} = \mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} + \mathcal{R}_{\sigma,k}$.

The parity and first-counterloop packets are typed by

$$\lambda_-(\sigma) = -(1 - \delta_{\sigma}), \quad \delta_{\sigma} = C_* \sqrt{\sigma} + o(\sqrt{\sigma}), \quad (13)$$

$$\beta = (r_H \sqrt{\lambda})^{-1}, \quad \beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right], \quad (14)$$

where $C_*, C_M > 0$, and

$$\mathcal{P}_{\sigma,2k} = r_H^{-2k} \{1 - \lambda_-(\sigma)^{2k}\} > 0, \quad (15)$$

$$\mathcal{A}_{k,2k} = (2k - 2)\beta_k^{2k} + 2\beta^{2k} > 0. \quad (16)$$

The exact signed coefficient ledger inherited from RH-326 and RH-327 is

$$\boxed{e_{\sigma,k} = \mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} + \mathcal{R}_{\sigma,k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k}}. \quad (17)$$

The target is

$$H_k = kR^{-2k}. \quad (18)$$

All moving-order limits use

$$k = k_{\sigma} = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_{\sigma} = k_{\sigma} - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad \eta_{\sigma} \longrightarrow \eta \quad (19)$$

along a fixed-phase subsequence. Thus

$$d_{\sigma,k} := \frac{\Delta_k^{\text{clr}}}{\sigma} \longrightarrow C_b \lambda^{-2\eta}. \quad (20)$$

For $d \geq 0$, let

$$\phi(t) = (2\pi)^{-1/2} e^{-t^2/2}, \quad \Phi(d) = \int_{-\infty}^d \phi(t) dt, \quad g_d(v) = \frac{\phi(v-d)}{\Phi(d)} \mathbf{1}_{v \geq 0}. \quad (21)$$

With standard normals Z_1, Z_2 independent of V and of each other, the retained local frame is

$$V \sim g_d, \quad U = -2u_c V - Z_1, \quad W = -\lambda U + Z_2, \quad (V, U, W) \text{ has orientation } (+, -, +), \quad (22)$$

with output shift $\kappa_{\text{aff}} d = 2u_c \lambda d$. These coordinates remain interface data; they do not identify the cyclic observation in (10).

2 The exact fixed-reference matching equation

Fix $L = L_{\sigma,k} > 0$ and a deterministic reference contrast $c_0 = c_{0,k} \in [-1, 1]$. For a proposed physical contrast $c_{\text{phys}} = c_{\sigma,k}^{\text{phys}} \in [-1, 1]$, put

$$X_{k,L}(c_{\text{phys}}; c_0) = L\{c_{\text{phys}}^{2k} - c_0^{2k}\}. \quad (23)$$

The actual shell slot has a representation error

$$\mathcal{E}_{\sigma,k}^{\text{obs}} := \mathcal{S}_{\sigma,k} - X_{k,L}(c_{\text{phys}}; c_0). \quad (24)$$

This is a definition once L, c_0, c_{phys} are supplied. No smallness or physical identification is assumed.

Define the signed demand and required even power by

$$D_{\sigma,k} = \mathcal{A}_{k,2k} - \mathcal{P}_{\sigma,2k} - \mathcal{B}_{\sigma,k}, \quad (25)$$

$$y_{\sigma,k} = c_0^{2k} + \frac{D_{\sigma,k}}{L}. \quad (26)$$

Theorem 2.1 (Exact fixed-reference joint equation). *For every typed choice above,*

$$\boxed{e_{\sigma,k} = L\{c_{\text{phys}}^{2k} - y_{\sigma,k}\} + \mathcal{E}_{\sigma,k}^{\text{obs}} + \mathcal{R}_{\sigma,k}}. \quad (27)$$

Moreover, the image of (23) over $|c_{\text{phys}}| \leq 1$ is

$$I_{k,L,c_0} = [-L|c_0|^{2k}, L(1 - |c_0|^{2k})], \quad (28)$$

and the RH-327 best-case screen has the exact normalized form

$$\text{dist}(D_{\sigma,k}, I_{k,L,c_0}) = L \text{dist}(y_{\sigma,k}, [0, 1]). \quad (29)$$

Proof. Substitute $\mathcal{S} = X_{k,L} + \mathcal{E}^{\text{obs}}$ into (17) and use $D = \mathcal{A} - \mathcal{P} - \mathcal{B}$:

$$e = X_{k,L} - D + \mathcal{E}^{\text{obs}} + \mathcal{R} = L(c_{\text{phys}}^{2k} - y) + \mathcal{E}^{\text{obs}} + \mathcal{R}.$$

Since c_{phys}^{2k} ranges exactly over $[0, 1]$, translation by $-Lc_0^{2k}$ proves (28). The affine change $D = L(y - c_0^{2k})$ then gives (29). $\square$

Corollary 2.2 (Conditional physical closure criterion). *For an identified physical representation, target closure is equivalent to the signed condition*

$$L(c_{\text{phys}}^{2k} - y_{\sigma,k}) + \mathcal{E}_{\sigma,k}^{\text{obs}} + \mathcal{R}_{\sigma,k} = o(H_k). \quad (30)$$

If additionally $\mathcal{E}_{\sigma,k}^{\text{obs}} + \mathcal{R}_{\sigma,k} = o(H_k)$, then it is equivalent to

$$c_{\text{phys}}^{2k} - y_{\sigma,k} = o(H_k/L). \quad (31)$$

This corollary is conditional on the physical fields. The repository does not identify L, c_0, c_{phys} , prove (24) small, or control $\mathcal{R}$.

3 Phase law and contrast conditioning

Normalize by the positive alias packet:

$$q_{\sigma,k} = \frac{\mathcal{P}_{\sigma,2k}}{\mathcal{A}_{k,2k}}, \quad b_{\sigma,k} = \frac{\mathcal{B}_{\sigma,k}}{\mathcal{A}_{k,2k}}, \quad \ell_{\sigma,k} = \frac{L}{\mathcal{A}_{k,2k}}, \quad z_{0,k} = c_0^{2k}. \quad (32)$$

Proposition 3.1 (Phase-normalized required power). *The required power is exactly*

$$y_{\sigma,k} = z_{0,k} + \frac{1 - q_{\sigma,k} - b_{\sigma,k}}{\ell_{\sigma,k}}. \quad (33)$$

On a fixed phase, RH-326 gives

$$q_{\sigma,k} = C_* C_M \lambda^\eta + o(1). \quad (34)$$

If also $b_{\sigma,k} \rightarrow b_\eta$, $\ell_{\sigma,k} \rightarrow \ell_\eta \in (0, \infty)$, and $z_{0,k} \rightarrow z_{0,\eta}$, then

$$y_{\sigma,k} \rightarrow y_\eta := z_{0,\eta} + \frac{1 - C_* C_M \lambda^\eta - b_\eta}{\ell_\eta}. \quad (35)$$

When $y_\eta \in (0, 1)$, the unique required contrast magnitude satisfies

$$r_k := y_{\sigma,k}^{1/(2k)} = 1 + \frac{\log y_\eta}{2k} + o(k^{-1}). \quad (36)$$

Proof. Divide (26) by the definitions in (32) to obtain (33). The fixed phase law yields (35). Finally, $r_k = \exp\{(\log y_{\sigma,k})/(2k)\}$ and $\log y_{\sigma,k} \rightarrow \log y_\eta$. $\square$

The limits of $b_{\sigma,k}$ and $\ell_{\sigma,k}$ are assumptions, not archived physical theorems. The same phase simultaneously fixes the clearance in (20); it cannot be optimized independently after inspecting (33).

Theorem 3.2 (Exponential contrast-radius precision). *Let $y_k \in [a, 1]$ for some fixed $a > 0$, let $\rho_k = |c_{\text{phys}}| \in [0, 1]$, put $r_k = y_k^{1/(2k)}$, and suppose $\varepsilon_k := H_k/L \rightarrow 0$. Then*

$$\rho_k^{2k} - y_k = o(\varepsilon_k) \iff \rho_k - r_k = o(\varepsilon_k/k). \quad (37)$$

If in addition

$$\mathcal{A}_{k,2k} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad \frac{L}{\mathcal{A}_{k,2k}} \rightarrow \ell \in (0, \infty), \quad (38)$$

then

$$\frac{H_k}{L} = \frac{C_M}{2\ell} (\beta R)^{-2k} \{1 + o(1)\}, \quad (39)$$

$$|\rho_k - r_k| = o\left(\frac{(\beta R)^{-2k}}{k}\right) \quad (40)$$

is the contrast-radius demand corresponding to (31).

Proof. Assume first that the power mismatch is $o(\varepsilon_k)$. For large k it is at most $a/2$, so both y_k and ρ_k^{2k} are at least $a/2$. The mean-value theorem gives

$$2k \min(\rho_k, r_k)^{2k-1} |\rho_k - r_k| \leq |\rho_k^{2k} - r_k^{2k}| \leq 2k |\rho_k - r_k|. \quad (41)$$

Because $0 \leq \min(\rho_k, r_k) \leq 1$ and its $2k$ th power is at least $a/2$, the left coefficient is at least ka . This proves the forward implication. The upper bound in (41) proves the converse. Finally, divide $H_k = kR^{-2k}$ by (38) to obtain (39); substitute into (37). $\square$

At even order the signs c_{phys} and $-c_{\text{phys}}$ are identical in every formula above. Thus even perfect matching identifies at most a contrast magnitude.

4 Sharp uncertainty certificate and a false-positive theorem

Retain the RH-325 trace-observation Duhamel ledger. Let $A_j, G_j : B_{j-1} \rightarrow B_j$, $B_m = B_0$, and suppose $|\mathfrak{t}(H)| \leq T\|H\|$. Then

$$|\mathfrak{t}(A_m \cdots A_1 - G_m \cdots G_1)| \leq \mathcal{U}_{\sigma,k} := \sum_{j=1}^m W_j \delta_j, \quad (42)$$

where

$$\delta_j = \|A_j - G_j\|, \quad W_j = T \prod_{\ell=j+1}^m \|A_\ell\| \prod_{\ell=1}^{j-1} \|G_\ell\|. \quad (43)$$

No Markov L^1 contraction may replace the observation norm or these prefix/suffix factors. Applying (42) to $\mathcal{E}^{\text{obs}}$ additionally requires an identified operator-product realization of the shell observation; no such realization is proved here.

Theorem 4.1 (Sharp symmetric uncertainty interval). *Put*

$$m_{\sigma,k} = L(c_{\text{phys}}^{2k} - y_{\sigma,k}). \quad (44)$$

If $|\mathcal{E}_{\sigma,k}^{\text{obs}}| \leq \mathcal{U}_{\sigma,k}$ and $|\mathcal{R}_{\sigma,k}| \leq V_{\sigma,k}$, then the set of residuals compatible with only these symmetric bounds is exactly

$$[m_{\sigma,k} - \mathcal{U}_{\sigma,k} - V_{\sigma,k}, m_{\sigma,k} + \mathcal{U}_{\sigma,k} + V_{\sigma,k}]. \quad (45)$$

Its best possible and worst possible absolute residuals are

$$E_{\text{best}} = \max\{|m_{\sigma,k}| - \mathcal{U}_{\sigma,k} - V_{\sigma,k}, 0\}, \quad (46)$$

$$E_{\text{worst}} = |m_{\sigma,k}| + \mathcal{U}_{\sigma,k} + V_{\sigma,k}. \quad (47)$$

Consequently every error pair allowed by the bounds gives $o(H_k)$ if and only if $E_{\text{worst}} = o(H_k)$.

Proof. Equation (27) is $e = m + \mathcal{E}^{\text{obs}} + \mathcal{R}$. The Minkowski sum of the two symmetric error intervals is $[-\mathcal{U} - V, \mathcal{U} + V]$, proving (45). Distance from zero to this interval gives (46); its farthest endpoint gives (47). $\square$

The best case is an existential statement over an uncertainty class. It does not show that the actual observation error and remainder choose the cancelling values.

Proposition 4.2 (Exact reachability can be a target-scale false positive). *Fix $0 < \theta < 1$, take $c_0 = 0$, and set $D_{\sigma,k} = \theta L$. Then*

$$\text{dist}(D_{\sigma,k}, I_{k,L,0}) = 0, \quad y_{\sigma,k} = \theta. \quad (48)$$

Nevertheless the admissible designated contrast $c_{\text{phys}} = 0$ with $\mathcal{E}^{\text{obs}} = \mathcal{R} = 0$ gives

$$e_{\sigma,k} = -\theta L. \quad (49)$$

Hence if $L/H_k \rightarrow \infty$, the best-case screen vanishes identically while $|e_{\sigma,k}|/H_k \rightarrow \infty$.

Proof. For $c_0 = 0$, (28) is $[0, L]$, which contains θL . Equation (26) gives $y = \theta$. Substitution of $c_{\text{phys}} = 0$ into (27) proves the residual formula. $\square$

This is a scalar information-class counterexample. It is not evidence that the actual noisy operator selects $c_{\text{phys}} = 0$ or that its full trace diverges.

5 Reproducible formula diagnostics

The code evaluates only proved formulas. For a visible phase diagnostic, take $k = 16$, $b_{\sigma,k} = 0.2$, $\ell_{\sigma,k} = 1$, and $c_0 = 0.8$. Using the archived ordinary floating-point constants gives

η	$C_* C_M \lambda^\eta$	$C_b \lambda^{-2\eta}$	y	$y^{1/(2k)}$
−1	0.122050	1.298369	0.678743	0.987963
−0.5	0.158127	0.773495	0.642665	0.986278
0	0.204869	0.460805	0.595923	0.983954
0.5	0.265428	0.274522	0.535364	0.980664
1	0.343888	0.163545	0.456904	0.975820

The boundary and shell ratios in this table are synthetic. They are not fitted or measured physical trace ratios.

With the leading alias scale $L = (2k/C_M)\beta^{2k}$ at phase zero, the matching and false-positive scales are

k	H_k/L	$H_k/(kL)$	false-positive $ e /H_k$ at $\theta = 1/4$
8	2.0911×10^{-2}	2.6138×10^{-3}	1.1956×10^1
16	4.4931×10^{-4}	2.8082×10^{-5}	5.5641×10^2
32	2.0744×10^{-7}	6.4826×10^{-9}	1.2052×10^6
64	4.4219×10^{-14}	6.9092×10^{-16}	5.6537×10^{12}

These are formula evaluations, not interval-certified constants or physical shell data. Finite rows are reproduction checks only and are not promoted to all-order noisy asymptotics.

6 Claim boundary and RH-329 handoff

The paper proves the exact conditional matching equation, its normalized phase law, the contrast-conditioning theorem, the sharp uncertainty interval, and a scoped reachability false positive. It does not prove that the actual shell has the exchange representation, identify its scale or contrast, control the RH-325 observation norm and every Duhamel weight, bound the second physical critical leg, prove $\mathcal{E}^{\text{obs}} = o(H_k)$ or $\mathcal{R} = o(H_k)$, or establish actual joint matching.

RH-329 should audit one fully fixed isolated model. It must freeze the reference before seeing the demand, measure rather than optimize the physical contrast, retain every observation weight, and report both (46) and (47) in units of H_k . A failure should be published as a scoped isolated-model negative result. A finite pass would still not identify the actual noisy operator or prove the RH-330 full-trace transfer criterion.

Gates A–E remain false/open. This work constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or completed-zeta divisor equality, and does not imply the Riemann Hypothesis.